

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-2

DURATION: 40 MINUTES

SUMMER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Euler's identity, which is $e^{i\pi} + 1 = 0$, is sometimes referred to as the most beautiful equation in mathematics. It is a special case of Euler's formula 6
(CO2)

$$e^{ix} = \cos(x) + i \sin(x) \quad \text{(PO1)}$$

when evaluated for $x = \pi$. One of the many proofs of this formula involves utilizing the Taylor expansions of its constituent transcendental functions.

The Taylor series for a function $f(x)$ is –

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of –

- (i) e^x
 (ii) $\cos(x)$
 (iii) $\sin(x)$

Then use these three series to prove Euler's formula.

2. Differentiate between *Interpolation* and *Extrapolation*. With suitable real-world examples, briefly write when each of the methods is applicable. 2
(CO3)
(PO1)
3. A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Table-1 portrays multiple recorded observations involving a thermistor. 4
(CO2)
(PO1)

Table 1: Temperature T of a thermistor as a function of its resistance R for Question-3.

Resistance, R (Ω)	Temperature, T ($^{\circ}C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the Lagrangian method of interpolation and a second order polynomial.

4. Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique. 3
(CO4)
(PO1)